COMP201 - Computer Systems & Programming, Fall 2025
Department of Computer Engineering, Koç University

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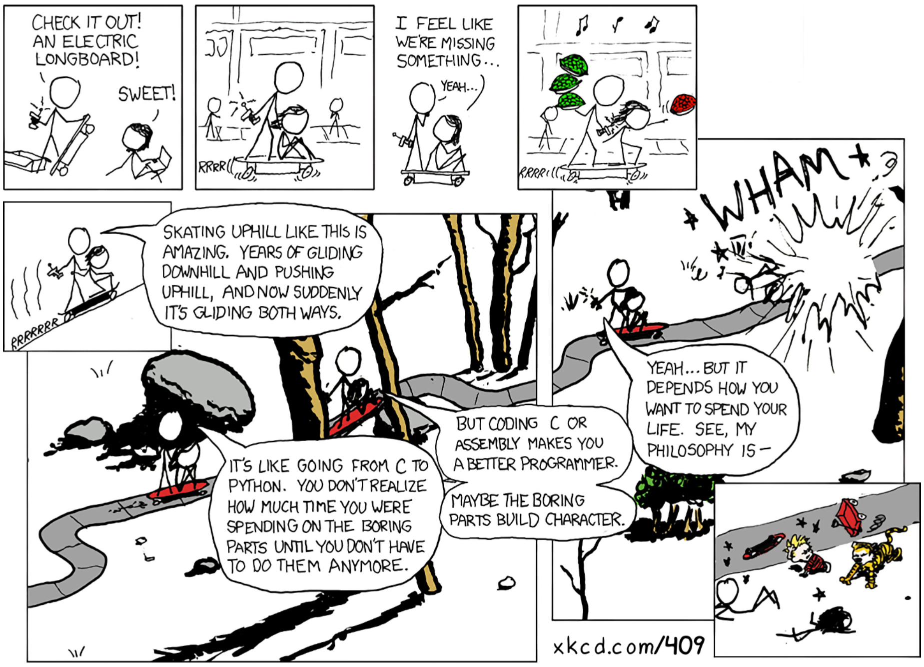
Lectures: Tueasday, Thursday at 16:00-17:10 (SNA B172).

Labs: Friday at 13:00-14:40 (Lab A) (SNAB242), 10:00-11:40 (Lab B) (SNA B242)



Electric Skateboard (Double Comic) · A web comic from xkcd, by Randall Munroe

COURSE DESCRIPTION

This course gives a solid understanding of the principles and abstractions used in computer systems and machine programs using C. Towards this aim, it covers a broad range of topics, providing students with an in-depth perspective and programming experience regarding the basic topics of C language and how programs are formed and executed at the microprocessor-level.

Upon the completion of COMP201, students will be able to

1. demonstrate proficiency in writing computer programs that require effective memory management,
2. gain a deep knowledge of the compilation flow and runtime behavior of C programs,
3. have a clear understanding of computer arithmetic in a modern computing system,
4. recognize the relationship between a computer program and its assembly translation, and
5. gain a general sense of working in a Unix environment as a power user, getting familiar with shell tools, version control systems, compilers, debuggers, and profilers.

Prerequisites

The prerequisite for COMP201 is COMP 132. You should have a solid programming background to appreciate the technical aspects regarding the relationship between C and Assembly programs.

Webpage The webpage is at `https://aykuterdem.github.io/classes/comp201.f25`.

Communication
The course webpage will be updated regularly throughout the semester with lecture notes, programming and reading assignments and important deadlines. All other communications will be carried out through KUHub Learn.

Reference Books

- Randal E. Bryant and David R. O'Hallaron, Computer Systems: A Programmer's Perspective, Third Edition, Pearson, 2016
- The C Programming Language, Kernighan and Ritchie
- Anish Athalye, Jon Gjengset and Jose Javier Gonzalez Ortiz, The Missing Semester of Your CS Education, MIT, 2020

Grading Policy
The grading for COMP201 will be based on

- 9 labs (your lowest 2 scores dropped, no make-up) (21%),
- class participation (5%),
- 5 programming assignments (18%) (Assignment0 2%, the others 4% each),
- a midterm exam (28%), and
- a final exam (28%).

Lectures
The lectures will be delivered face to face, and will be recorded and be made available via KUHub Learn. **The students are expected to attend at least 70% of the lectures**, and they are strongly encouraged to be actively participate in class discussions and interact with the instructor and the other students. You will be held responsible for all material presented at the lectures.

Labs
This semester there will be **in total 9 labs**. While lectures will mostly introduce high-level concepts, these labs will supply practical sessions to familiarize you with the Unix/Linux development tools and give you hands-on experiences in the context of course topics. **Each lab will have two graded parts.** The first one will be a short prelab assignment that you need to complete and submit **before the lab session**. TThe second part will be a practice problem to be completed **during the lab session** and submitted by the end of it. Please note that **late submissions are not allowed** for either part. Also note that **your lowest 2 scores will be dropped, hence there will be no make-up** for the lab sessions. Like the lectures, the labs will be in person.

Assignments
There will be **1 warm-up and 4 programming assignments.** You are encouraged to discuss with your classmates about the given assignments, but these discussions should be carried out in an abstract way. **All work on assignments has to be done individually.** In short, turning in someone else's work, in whole or in part, as your own will be considered as *a violation of academic integrity*. Please note that the former condition also holds for the material found on the web, as everything on the web has been written by someone else.

You may use up to 7 *grace days* (in total) over the semester for the assignments. That is, you can submit your solutions without any penalty if you have free grace days left. Any additional unapproved late submission will be punished (1 day late: 20% off, 2 days late: 40% off) and **no submission 2 days after the original deadline will be accepted (grace days included!).**

Exams
There are a midterm exam and a final exam, which will include mostly open-ended fill-in-the-blanks or short answer type questions. **Please note that you can only take a make-up for either the midterm or the final exam, but <u>not both</u>.**

Good luck, and have fun!

Tentative Schedule

W	*Date*	*Topic*	*Notes*
1	10/7	Introduction, Course logistics, A tour of C programs	B&O 1
	10/9	Bits & Bytes, Representing and Operating on Integers	B&O 2.2-2.3
	10/10	**Bootcamp:** Programming with C and Git basics	
2	10/14	Bits and Bitwise Operators	B&O 2.1
	10/16	Floating point	B&O 2.4
		Assg0 out: *Getting Started with Unix and C*	
	10/17	**Lab 1:** The Linux Shell	
3	10/21	Chars and Strings in C	K&R 1.9, 5.5, Appx B3
	10/23	More Strings, Pointers	K&R 1.6, 5.5, Essential C 3
		Assg0 in, Assg1 out: Strings in C	
	10/24	**Lab 2:** Bits, Ints and Floats, Vim	MIT MS Editors (Vim)
4	10/28	*No classes – Republic Day*	
	10/30	Arrays and Pointers	K&R 5.2-5.5, Essential C 6
	10/31	**Lab 3:** Strings in C and GDB	
5	11/4	The Stack and The Heap	K&R 5.6-5.9, Essential C 6
	11/6	Realloc, Memory Bugs	K&R 5.6-5.9, Essential C 6
		Assg1 in	
	11/7	**Lab 4:** Arrays, Pointers, and Valgrind	
6	11/11	`void *`, Generics	K&R 5.6-5.9, Essential C 6
	11/13	Function Pointers	K&R 5.11
	11/14	*No labs this week*	
7	11/18	`const`, Structures	K&R 6.1-6.7
	11/20	Midterm Review	
		Midterm Exam	
	11/21	*No lab this week*	
8	11/25	Compiling C programs	Stanford Unix Programming Tools 1
		Assg2 out: Heap Management	
	11/27	Intro to x86-64, Data Movement	B&O 3.1-3.4
	11/28	**Lab 5:** Structs, Working with multiple files, writing Makefiles	

W	*Date*	*Topic*	*Notes*
9	12/2	Arithmetic and Logic Operations	B&O 3.5-3.6
	12/4	x86-64 Control Flow	B&O 3.6.1-3.6.2
	12/5	**Lab 6:** Machine Programming with Assembly	
10	12/9	More Control Flow	B&O 3.6.3-3.6.8
		Assg2 in, **Assg3 out:** Defusing a Binary Bomb	
	12/11	x86-64 Procedures	B&O 3.7
	12/12	*No labs this week*	
11	12/16	Security Vulnerabilities	B&O 3.10
	12/18	Data and Stack Frames	B&O 3.8-3.9
		Assg3 in, Assg4 out: Buffer Overflow Bugs	
	12/19	**Lab 7:** Runtime Stack	
12	12/23	Cache Memories	B&O 6.1-6.4.2
	12/25	More Cache Memories	B&O 6.4-6.7
	12/26	**Lab 8:** Memory organization	
13	12/30	Optimization	B&O 5
	1/1	*No classses – New Year's Day*	
		Assg4 in	
	1/3	*No labs this week*	
14	1/6	Linking	B&O 7
	1/8	Wrapping Up	
	1/9	**Lab 9:** Code optimization	